Mini Project Report

on

BondHealth

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In partial fulfilment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering.

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DIVISION OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

Certified that this is a Mini Project Report titled

BondHealth

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DECLARATION

We, Mariyam Rahim, Pillai Anjita, Sneha Vijayan, Soya R F and Thara Asharaf, hereby declare that the project titled **"BondHealth"** is a bona fide work carried out by us during the year 2026, under the guidance of **Mrs. Sheena S**, Assistant Professor, School of Engineering, CUSAT. We further declare that this project has not been submitted, either in part or full, for the award of any degree, diploma, associateship, fellowship, or any other academic recognition in any university or institution.

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ABSTRACT

BondHealth is a comprehensive digital healthcare ecosystem designed to unify patient care across multiple healthcare providers. This project addresses the critical challenge of fragmented medical records by creating a centralized platform where patients' medical history, appointments, and lab results are securely accessible across all connected hospitals.

The system features role-based dashboards for patients, doctors, hospital administrators, and lab technicians. Key functionalities include appointment scheduling, electronic medical records management, lab report integration, prescription management, and a public review system. Built with Node.js and Express on the backend, PostgreSQL for data persistence, and vanilla JavaScript with Tailwind CSS for the frontend, BondHealth demonstrates a production-ready healthcare management solution.

The platform implements robust authentication using JWT tokens, role-based access control, and comprehensive API endpoints for all operations. With features like real-time appointment management, leave tracking for doctors, and a dynamic review system, BondHealth provides a seamless experience for all stakeholders in the healthcare delivery chain.

Keywords: Healthcare Management System, Electronic Medical Records, Patient Portal, Telemedicine, Hospital Management, Node.js, PostgreSQL, JWT Authentication, Role-Based Access Control

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Chapter 1

Introduction

The healthcare industry today faces significant challenges in managing and accessing patient medical information. In the current system, patient data is often scattered across multiple hospitals and healthcare providers, creating fragmented medical histories that are difficult to access when needed. This fragmentation leads to several critical issues: patients become dependent on physical reports, undergo repeated diagnostic tests, face limited options for remote consultations, and encounter difficulties in obtaining certain medications, especially rare ones. Additionally, hospitals operate without a common digital platform to efficiently manage patient data and streamline interactions between healthcare providers and patients.

The digital transformation of healthcare has become increasingly important in recent years, with the COVID-19 pandemic highlighting the urgent need for remote healthcare solutions. Patients, particularly the elderly and those with mobility issues, require accessible healthcare services that do not always necessitate physical hospital visits. Furthermore, the need for a centralized, secure system to store and access medical records has become paramount in ensuring continuity of care and improving healthcare outcomes.

1.1 Problem Statement

The existing healthcare ecosystem suffers from the following key problems:

- **Fragmented Medical Records**: Patient medical information is distributed across different hospitals, making it difficult to obtain complete medical histories when required for diagnosis or treatment.
- **Dependency on Physical Reports**: Patients must carry physical copies of their medical reports, which can be lost, damaged, or forgotten during emergencies.
- **Repeated Diagnostic Tests**: Due to inaccessible medical histories, patients often undergo redundant tests, increasing healthcare costs and inconvenience.
- **Limited Remote Consultation Options**: Many patients, especially the elderly and those in remote areas, struggle to access healthcare services that require physical presence.
- **Difficulty in Obtaining Medications**: Patients face challenges in locating and purchasing rare or specific medicines that may only be available at certain hospital pharmacies.

- **Lack of a Common Digital Platform**: Hospitals lack a unified digital interface to efficiently manage patient data, appointments, consultations, and billing processes.

1.2 Objectives

The primary objectives of the BondHealth project are:

1. **Develop a Centralized Digital Platform**: Create a web-based healthcare platform that connects hospitals and patients within a specific geographical area, functioning as a digital interface for healthcare services.

2. **Implement Digital Patient Identities**: Provide unique, secure digital IDs to patients (similar to DigiLocker) for storing and accessing medical records including consultation history, diagnostic reports, and medical scans.

3. **Enable Hospital Discovery**: Display registered hospitals along with available departments and doctor information, allowing patients to make informed decisions about their healthcare providers.

4. **Facilitate Remote Consultations**: Support scheduled chat and video call consultations between patients and doctors, making healthcare accessible to those unable to visit hospitals in person.

5. **Integrate Pharmacy Services**: Enable online purchase of medicines, including rare medications, from hospital pharmacies with electronic billing for transparent, paperless transactions.

6. **Ensure Security and Privacy**: Implement robust security measures to protect sensitive patient data and ensure compliance with healthcare data protection standards.

1.3 Scope of the Project

The scope of BondHealth encompasses the following aspects:

1.3.1 Geographical Scope

The platform initially targets a specific geographical area, connecting hospitals and patients within that region. This focused approach allows for better management, testing, and refinement of the system before potential expansion to wider areas.

1.3.2 Functional Scope

- **Patient Registration and Authentication**: Secure registration process assigning unique patient IDs with robust authentication mechanisms.
- **Medical Records Management**: Digital storage and retrieval of consultation histories, diagnostic reports, and medical scans.
- **Hospital and Doctor Directory**: Comprehensive listing of registered hospitals with department-wise doctor information.
- **Teleconsultation Services**: Scheduling and conducting remote consultations through chat and video calls.
- **Online Pharmacy**: E-commerce functionality for medicine purchases from hospital pharmacies.
- **Electronic Billing**: Digital invoice generation and payment processing for consultations and medicine purchases.

1.3.3 Technical Scope

The project involves developing a responsive web application using modern web technologies, implementing secure database management for medical records, integrating video calling APIs for teleconsultation, and ensuring data encryption and privacy compliance.

Chapter 2

Literature Review

The rapid growth of digital technologies has significantly transformed modern healthcare systems. Hospitals are increasingly adopting digital platforms to improve operational efficiency, enhance patient engagement, and strengthen communication between patients and healthcare providers. Digital hospital systems, telemedicine platforms, and patient-centered health technologies play a major role in improving healthcare accessibility, quality of care, and overall patient satisfaction. This chapter reviews recent literature related to digital hospital systems, telemedicine, digital patient engagement, and patient-provider relationships.

The technologies used in digital hospitals include cloud-based platforms, integrated Electronic Medical Record (EMR) systems, mobile health applications, AI-based analytics, and secure hospital information systems. However, the study also identifies limitations such as high implementation cost, resistance from healthcare staff, technical complexity, data privacy concerns, and increased cognitive workload for clinicians. In some cases, excessive digital documentation reduced face-to-face interaction time with patients.

2.1 Digital Hospital Systems and Patient Experience

This study analyzes how fully digital hospitals influence both patient and clinician experiences. The functionality of digital hospital systems includes Electronic Health Records (EHR), digital medication management, clinical decision support systems, patient portals, and automated workflow systems. These systems aim to improve care coordination, reduce medical errors, enhance documentation accuracy, and enable real-time access to patient data. The study emphasizes improved communication, streamlined workflows, and enhanced patient safety as key outcomes of digital hospital implementation. Technologies used in digital hospitals include cloud-based platforms, integrated Electronic Medical Record (EMR) systems, mobile health applications, AI-based analytics, and secure hospital information systems. Research emphasizes improved communication, streamlined workflows, and enhanced patient safety as key outcomes of digital hospital implementation. However, limitations include high implementation costs, resistance from healthcare staff, technical complexity, data privacy concerns, and increased cognitive workload for clinicians. In some cases, excessive digital documentation reduces face-to-face interaction time with patients.

2.2 Telemedicine and Digital Patient Engagement

This research focuses on redesigning telemedicine models to improve accessibility and patient engagement. The functionality of the proposed assisted telemedicine model includes

remote consultations, virtual diagnosis support, digital appointment scheduling, remote patient monitoring, and digital follow-up care. The model particularly supports rural and underserved populations by integrating assisted telehealth centers where healthcare workers help patients access digital consultations.

The technologies used include video conferencing platforms, remote diagnostic devices, cloud-based health record storage, wearable health monitoring devices, and secure communication protocols. Despite its advantages, the system faces limitations such as internet connectivity issues, digital literacy barriers among patients, regulatory challenges, and limited physical examination capabilities. Additionally, data security and patient privacy remain major concerns.

2.3 Digital Health Technologies in Patient Experience

This scoping review examines various digital health technologies that influence patient experience. The functionality of these technologies includes mobile health (mHealth) apps, AI-driven chatbots, online appointment systems, digital feedback systems, and patient engagement portals. These tools aim to increase patient participation in treatment decisions, improve transparency in healthcare services, and provide personalized health recommendations.

The technologies discussed in the review include Artificial Intelligence (AI), Machine Learning (ML), wearable IoT devices, blockchain for secure data sharing, and mobile-based applications. However, the review highlights limitations such as technology accessibility gaps, interoperability issues between different hospital systems, ethical concerns in AI-based decision-making, and unequal access among elderly or low-income populations. Furthermore, long-term effectiveness data is still limited.

2.4 Telemedicine and Patient-Provider Relationships

This study explores how telemedicine impacts patient-provider relationships. The functionality of telemedicine platforms includes video consultations, secure messaging, digital prescription systems, and online medical record sharing. These platforms enhance convenience, reduce travel time, and allow continuous patient monitoring. The study highlights improved access to healthcare services and flexibility in appointment scheduling.

The technologies used include secure teleconferencing systems, encrypted communication channels, digital prescription software, and cloud-based data management systems. However, limitations include reduced physical interaction, potential miscommunication, technical glitches during consultations, and challenges in building trust through virtual platforms. In some cases, lack of emotional connection and non-verbal cues may affect patient satisfaction.

2.5 Summary

The proposed Bond-Health Hospital–Patient Interface System aims to overcome these limitations by providing:

- A centralized digital platform connecting hospitals and patients
- Simplified appointment and specialist management
- Secure patient data handling
- A user-friendly interface for both hospital administrators and patients
- Improved communication between healthcare providers and patients

Unlike many complex digital hospital systems, Bond-Health focuses on simplicity, accessibility, and integrated management of hospital services such as doctor registration, specialty management, laboratory, and pharmacy modules. By combining essential digital functionalities with an easy-to-use interface, the system enhances patient experience while maintaining operational efficiency.

Thus, this project contributes to the growing field of digital healthcare by offering a practical, scalable, and patient-centered hospital interface solution.

Chapter 3

SYSTEM STUDY

The system study phase involves understanding the existing healthcare management challenges and analyzing the requirements for BondHealth. This chapter covers the problem definition, feasibility analysis, and detailed requirements gathering for the proposed system.

3.1 Existing System

3.1.1 Current Healthcare Challenges

- **Fragmented Medical Records:** Patient data is scattered across different hospitals, clinics, and laboratories with no central access.
- **Manual Appointment Management:** Many healthcare facilities still rely on phone calls and paper-based appointment scheduling.
- **Delayed Lab Results:** Test reports take days to reach doctors and patients due to manual delivery systems.
- **Lack of Patient Engagement:** Patients have limited access to their own health information and medical history.
- **Administrative Inefficiency:** Hospital staff spend excessive time on paperwork and manual data entry.

3.1.2 Disadvantages of Existing System

- No centralized patient health record system
- Increased wait times for appointments and results
- Higher chances of medical errors due to missing information
- Poor communication between different healthcare providers
- Limited scalability for growing healthcare networks

3.2 Proposed System

3.2.1 System Overview

BondHealth is a web-based healthcare management platform that connects patients, doctors, hospitals, and laboratories on a single unified system. The platform provides:

- **Centralized Patient Records:** Complete medical history accessible to authorized healthcare providers
- **Online Appointment System:** 24/7 appointment booking with real-time availability
- **Digital Lab Reports:** Instant sharing of test results between labs, doctors, and patients
- **Role-Based Dashboards:** Customized interfaces for different user types
- **Secure Authentication:** JWT-based authentication with role-based access control

3.2.2 Advantages of Proposed System

- Seamless access to patient medical history across all connected hospitals
- Reduced wait times through online appointment scheduling
- Instant delivery of lab reports to doctors and patients
- Enhanced patient engagement through personal health portal
- Improved administrative efficiency with automated workflows
- Scalable architecture for future expansion

3.3 Feasibility Study

3.3.1 Technical Feasibility

The system is technically feasible as it uses:

- **Node.js & Express:** Proven backend technology for building scalable web applications
- **PostgreSQL:** Robust relational database with excellent performance and data integrity
- **JWT Authentication:** Industry standard for secure API authentication
- **Responsive Design:** Works seamlessly on desktops, tablets, and mobile devices

3.3.2 Operational Feasibility

The system is operationally feasible because:

- User-friendly interfaces require minimal training for staff
- Role-based dashboards simplify daily tasks for each user type
- Automated notifications reduce manual follow-up work
- Centralized records eliminate redundant data entry

3.3.3 Economic Feasibility

The system offers significant economic benefits:

- Reduced administrative costs through automation
- Elimination of paper-based systems and storage
- Improved staff productivity and efficiency
- Scalable architecture grows with organizational needs
- Open-source technology stack minimizes licensing costs

3.4 Requirements Analysis

3.4.1 Functional Requirements

3.4.1.1 Patient Module

- User registration and profile management
- View and manage appointments
- Book new appointments with available doctors
- Access medical reports and prescriptions
- Update personal and emergency contact information
- Write and view public reviews

3.4.1.2 Doctor Module

- View today's appointment schedule
- Access patient medical history
- Review and add findings to lab reports
- Manage availability and leave requests
- Communicate with patients

3.4.1.3 Hospital Admin Module

- Register new doctors and staff
- Manage medicine inventory
- Add laboratory facilities
- View hospital statistics and reports
- Manage doctor leave schedules

3.4.1.4 Lab Technician Module

- Upload and manage lab reports
- Update report status (pending/completed)
- Share results with doctors and patients
- View pending test requests

3.4.2 Non-Functional Requirements

3.4.2.1 Performance Requirements

- Page load time should be less than 3 seconds
- System should handle 1000+ concurrent users
- Database queries should execute within 500ms
- API response time under 200ms for standard operations

3.4.2.2 Security Requirements

- All passwords must be hashed using bcrypt
- JWT tokens for API authentication with 7-day expiration
- Role-based access control for all operations
- Session management with active session tracking
- Secure cookie handling with HttpOnly flag

3.4.2.3 Usability Requirements

- Intuitive interface requiring minimal training
- Responsive design for all device sizes
- Clear error messages and user feedback
- Consistent navigation across all modules

3.4.2.4 Reliability Requirements

- 99.9% uptime during business hours
- Data backup and recovery procedures
- Transaction integrity for critical operations
- Graceful error handling without data loss

3.5 System Architecture Overview

Three-Tier Architecture

- **Presentation Layer:** HTML, CSS, JavaScript, Tailwind CSS
- **Application Layer:** Node.js, Express.js
- **Data Layer:** PostgreSQL Database

Figure 3.1: BondHealth Three-Tier Architecture

The BondHealth system follows a three-tier architecture:

1. **Client Tier:** Web browsers render HTML pages with dynamic content fetched via API calls

2. **Application Tier:** Express.js server handles HTTP requests, implements business logic, and manages authentication
3. **Data Tier:** PostgreSQL database stores all application data with proper relationships and constraints

3.6 System Modules

3.6.1 Authentication Module

Handles user registration, login, and session management. Implements:

- JWT token generation and verification
- Password hashing with bcrypt
- Role-based access control
- Session tracking and management

3.6.2 Patient Management Module

Manages all patient-related operations:

- Profile creation and updates
- Medical history tracking
- Appointment scheduling
- Report and prescription access

3.6.3 Appointment Management Module

Handles the complete appointment lifecycle:

- Real-time availability checking
- Appointment booking, rescheduling, and cancellation
- Conflict detection and prevention
- Status tracking and notifications

3.6.4 Reporting Module

Manages lab reports and medical documents:

- Report upload and storage
- Status tracking (pending/reviewed)
- Secure sharing between stakeholders
- Historical data access

3.7 User Roles and Permissions

Role	**Permissions**
Patient	View own records, book appointments, access reports
Doctor	View appointments, access patient history, review reports
Hospital Admin	Manage doctors, staff, inventory, view analytics
Lab Technician	Upload reports, update status, share results

Table 3.1: User Roles and Access Levels

3.8 Summary

The system study phase has established a clear understanding of the existing healthcare challenges and defined comprehensive requirements for BondHealth. The proposed system addresses critical gaps in patient data management, appointment scheduling, and report sharing. With its modular architecture, role-based access control, and focus on usability, BondHealth is well-positioned to transform healthcare delivery in connected facilities.